



OMKAR DESHMUKH · AI × HEALTHCARE

CASE FILE — OMKAR DESHMUKH · BENGALURU, IN

SOFTWARE ENGINEER

RESEARCHER · INNOVATOR ·
DEVELOPER & EVENT ORGANIZER

Building at the intersection of **artificial intelligence, biomedical engineering, and healthcare innovation** — from patented 3D bioprinting systems to AI radiology platforms, I make technology that treats real problems.

[View my work](#)[Get in touch](#)

7

PATENTS FILED

4

GRANTED · IN + US

₹17 Cr

IP VALUATION

₹30.45L

RESEARCH FUNDING

PLE DEV CENTER + 15.8L IMPRESSIONS + 3D BIOPRINTING + AI RADIOLOGY +

FILE 01 • PROFILE

Research-driven engineer with a *clinical* focus

I'm a motivated, research-oriented engineer with a strong interdisciplinary focus spanning **artificial intelligence, healthcare innovation, biomedical engineering, robotics, data analytics, and public policy**. My work is driven by one question: how can technology meaningfully improve health outcomes and solve societal challenges at scale?

That question has taken me from publishing research on CNN-based dermatological treatment systems, to filing **7 patents — 4 granted across India and the United States** — to leading Team Vidyut in **ISRO's STRC 2026** challenge, to founding and licensing **TEDxSNPSU**. I work across the full innovation pipeline — research, prototyping, intellectual property, and technology commercialization.

I'm actively involved in national and international research programmes, including fellowship and innovation initiatives at the **National University of Singapore**.

2024 — 2028 (EXPECTED)

B.E. Computer Science & Engineering

Sapthagiri NPS University, Bengaluru

NUS • SINGAPORE

Young Fellowship — FIERD Program

National University of Singapore

Selected from 2,68,483 applicants

CLASS XII

Higher Secondary Education

Maharashtra State Board, India

Secondary Education

CBSE, Maharashtra, India

FILE 02 • RESEARCH RECORD — 8 ENTRIES

Published work in AI and *biomedical* systems

Q2 RESEARCH PAPER

CNN Technologies for 3-D Bioprinting

Published Q2 research paper on convolutional neural network architectures applied to automated 3D bioprinting workflows for skin regeneration and dermatological treatment.

Q3 RESEARCH PAPER

Integrating AI Models on Technological Benchmarks

Published Q3 research paper on benchmarking and integration strategies for AI models across technological platforms.

CONFERENCE • IISC

Methods to Train U-Net, RNN and CNN Models

Presented at a national conference hosted by the Indian Institute of Science (IISc), Bengaluru.

CONFERENCE • DELHI UNIV.

Mathematical Representation of CNN Formulae

Presented at an international conference hosted by Delhi University.

Contributed chapter examining AI-driven optimization, forecasting, and decision systems across modern logistics and supply chain operations.

BOOK CHAPTER • ELSEVIER / SCOPUS

AI in Predictive Modelling and Diagnostic Systems

Contributed chapter on machine-learning approaches to predictive modelling and AI-assisted diagnostics in clinical settings.

BOOK CHAPTER • SPRINGER

Integrating AI for Big Data Models

Book chapter submitted to Springer on AI integration strategies for large-scale data modelling.

BOOK CHAPTER

AI Technologies for Dermatology

Book chapter submitted on applied AI technologies for dermatological diagnosis and treatment.

FILE 03 • INTELLECTUAL PROPERTY – 7 FILED / 4 GRANTED

An IP portfolio in AI-powered healthcare

7

PATENTS FILED

4

PATENTS
GRANTED

1

PATENT
PUBLISHED

1

COPYRIGHT
GRANTED

₹17 Cr

Integrated 3D Bioprinting Device for Dermatological Treatment

GRANTED ×4

Flagship innovation with **four granted patents** — **two in India and two in the United States**: an AI-integrated bioprinting device designed for dermatological applications including skin regeneration therapy — combining hardware, CNN-driven imaging, and automated clinical workflows. Patent design valuation assessed at approximately **₹17 Crore** through WIPO-based international IP valuation resources.

2× Granted — India 2× Granted — United States ₹17 Cr Valuation

3D Bioprinting CNN / U-Net Dermatology

CNN Model for Facial Augmentation

PUBLISHED

Indian patent filed and published, covering a convolutional neural network model for facial augmentation, recognition, and treatment-oriented analysis in clinical applications.

Patent Published — India CNN Computer Vision

AI-Powered Diagnostic Model for Radiology

© COPYRIGHT

Granted copyright protecting an AI-powered diagnostic model for radiological assessment and early disease detection.

Copyright Granted Medical Imaging Diagnostics

7 patents filed

Across AI, healthcare technologies, and biomedical engineering.

WIPO valuation

International patent valuation assessment through WIPO resources.

Technology validation, evaluation, and commercialization-oriented IP assessment.

Hands-on patent drafting and intellectual property strategy.

FILE 04 · BUILD INDEX — 7 PROJECTS

Selected builds — from *bioprinters* to space robotics

P-01

Automated 3D Bioprinter for Dermatology (Vitiligo)

AI-integrated bioprinting system using CNN/U-Net architectures for skin re-generation, workflow automation, and clinical decision support in vitiligo treatment.

PATENTED ×4 · CNN / U-NET
₹17 CR VALUATION

P-02

AI Radiological Assessment Platform

AI-powered healthcare platform for early disease detection and diagnostic assistance from radiological imaging.

COPYRIGHT · MEDICAL IMAGING
DEEP LEARNING

P-03

Machine Learning for Space Applications

Built predictive machine-learning models for space applications as Team Leader of Team Vidyut in the national-level ISRO STRC 2026 challenge.

ISRO STRC 2026 · TEAM VIDYUT
PREDICTIVE ML

P-04

ment-oriented diagnostic applications.

PATENT PUBLISHED · CNN
COMPUTER VISION

P - 05

3D Bioprinting Web Application

AI-assisted web application for print-path planning, imaging, and printer parameter configuration.

REACT · NODE.JS · REST APIS

P - 06

BioBridge — Biomedical Engineering Club Website

React-based platform for BioBridge, the cross-domain biomedical club I lead as President — supporting healthcare R&D initiatives on campus.

REACT · FRONTEND

P - 07

National-Level Drone Prototype

Designed and developed a drone prototype shortlisted for national-level competitions.

ROBOTICS · PROTOTYPING

FILE 05 · LEADERSHIP LOG

Leading teams, events, and communities

10+ EVENTS ORGANIZED TILL DATE

TEDxSNPSU — Dec 2025

TEDxSNPSU — May 2026

2× TEDx Awareness Pre-events

Research Methodology Workshop

Cross-Disciplinary Poster Presentation — Engg × Medical



JUL 2025 — PRESENT

Organizer & Licensee — TEDxSNPSU

Obtained the TED license and built TEDxSNPSU from the ground up: organized two TEDx events (December 2025 and May 2026) plus two awareness pre-events, managing speaker coordination, sponsorship outreach, and full event execution. Led digital campaigns that generated **~15.8 lakh impressions in 30 days**.



SNPSU MILESTONE

Established Apple Development Center at SNPSU

Drove the establishment of an Apple Development Center on campus — making SNPSU among the earliest universities in India to host one — and championed industry-academia collaboration, incubation, and startup culture in the university ecosystem.



OCT 2024 — 2026

Team Leader — ISRO STRC 2026 (Team Vidyut)

Leading **Team Vidyut** in the national-level ISRO STRC 2026 challenge — managing machine-learning development, collaboration, planning, and technical milestones.



SNPSU

President — BioBridge (Biomedical Engineering Club)

Founded and lead BioBridge, a cross-domain biomedical engineering club uniting students from medicine, engineering, and applied sciences — with a mission to promote healthcare R&D, interdisciplinary collaboration, and student-led innovation.



OCT 2024 — PRESENT

Robotics Director — Build Club (R&I Cell, SNPSU)

Spearheaded robotics projects and mentored students across the university's research and innovation cell — organizing the Robotics Challenge, Future Innovator Challenge, BIS Awareness Camp, a Research Methodology



SCSCD

Team Lead — District-Level Science & Football Competitions

Led planning and execution; secured prizes in multiple events. Also hosted the SCSCD Annual Day, winning first prize in event hosting.

FILE 06 • HONORS — 11 CITATIONS

National and international *honors*

Research Excellence

03 National Excellence Award

Youth R&D

05 Student Excellence Award 2026–27

SNPSU

07 Tracker of the Year

National-Level TCS Hackathon, Pune

09 1st Prize — Event Hosting & Science Competition

SCSCD

11 3rd Prize — Science Competition

District Level

04 Junior Scientist Honor

Worldwide Foundation

06 Outstanding Achievement in AI

NUS Data Science Conference

08 Best Use of AI

Presidency University, Bengaluru

10 1st Prize — Football & Debate

District Level

FILE 07 • FUNDING & FELLOWSHIPS

Government-funded *research*

₹18,00,000

Ministry of State Health & Family Welfare

Healthcare innovation and biomedical research initiatives, including Government of Maharashtra funding support for the 3-D Bioprinting for Dermatological Treatment project.

₹12,45,000

Ministry of Corporate Affairs

Prototype development, research, publications, and project operations.

FUNDING PROPOSAL PRESENTATIONS

Rajiv Gandhi Science & Technology Commission

Presented funding proposal at Nagpur.

ICMR

Second proposal presentation to the Indian Council of Medical Research.

Ministry of State Health & Family Welfare

Proposal presented — funded at ₹18,00,000.

Ministry of Corporate Affairs

Proposal presented — funded at ₹12,45,000.

SELECTED 1 OUT OF 2,68,483 STUDENTS

International exposure: chosen from over **2.68 lakh applicants** for fellowship and innovation programmes at the **National University of Singapore (NUS)**, including selection into the **FIERD Program** — with ongoing participation in national and international research collaborations across healthcare, engineering, and technology.

FILE 08 • TOOLKIT

Tools of the *trade*

AI / MACHINE LEARNING

Deep Learning

CNN / U-Net

LANGUAGES

Python

JavaScript

C

C#

Predictive Modeling

LLMS

Prompt Engineering

BACKEND & DATA

Node.js

REST APIs

MongoDB

SQL

Data Modelling

NumPy

Pandas

TOOLS & PRACTICE

Git / GitHub

React

Research Methodology

Innovation Management

Tech Commercialization

Technical Documentation

FILE 09 · LANGUAGES — 6 SPOKEN

Communicating across six *languages*

Marathi

MOTHER TONGUE

NATIVE

English

PROFICIENT

Hindi

PROFICIENT

Sanskrit

PROFICIENT

Kannada

INTERMEDIATE

Gujarati

INTERMEDIATE

FILE 10 • CERTIFICATIONS – 36+

35+ *certifications* across AI, cloud, data & security

Microsoft

15 CERTS

AI & AZURE OPENAI

Azure AI Fundamentals

AI Builder

Computer Vision

NLP

Azure OpenAI

Generative AI

Prompt Engineering

AI Agents

AI Applications

DATA & CLOUD

SQL

Power BI

Azure Data Fundamentals

Azure Fundamentals

Azure Administrator

Azure Developer

IBM

9 CERTS

ARTIFICIAL INTELLIGENCE

AI Fundamentals

AI Analyst

Generative AI

AI Ethics

AI Agents

Responsible AI

MACHINE LEARNING

Machine Learning Basics

Deep Learning

Neural Networks

Infosys Springboard

7+ CERTS

PROGRAMMING & AUTOMATION

Python Data Analyst

Python Automation

Enterprise Automation with Python

Database Management Systems

SECURITY & PROFESSIONAL

Cisco Networking Academy

5 CERTS

NETWORKING & SECURITY

Networking Basics

Introduction to Networks

Network Support

Packet Tracer

Introduction to Cybersecurity

OMKAR.D

ABOUT

RESEARCH

PATENTS

PROJECTS

LEADERSHIP

AWARDS

Cyber Security & Cyber Law Ethics

Soft Skills Learning

Plus **13 completed Python courses** and ongoing training programmes in AI, machine learning, and innovation management.

FILE 11 • CONTACT

Let's build something that **matters**.

Open to research collaborations, healthcare innovation projects, speaking engagements, and interesting problems worth solving.

Email me

Download CV ↓

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BASE Bengaluru, Karnataka · Native: Omerga, Maharashtra

